Familial Infantile Myoclonic Epilepsy (FIME) — Comprehensive Disease Report

Disease: Familial Infantile Myoclonic Epilepsy (FIME) **MONDO ID:** MONDO:0011506 · **OMIM:** 605021 · **Causal gene:** *TBC1D24* (16p13.3) **Category:** Mendelian, autosomal recessive

Summary

Familial Infantile Myoclonic Epilepsy (FIME) is a rare, autosomal-recessive, infantile-onset epilepsy caused by biallelic (loss-of-function or hypomorphic missense) variants in *TBC1D24* on chromosome 16p13.3. It is defined clinically by early-infantile myoclonic seizures accompanied by febrile convulsions and generalized tonic-clonic seizures, with generally preserved — or only mildly affected — cognitive development. FIME represents the **mild/benign end** of a strikingly broad *TBC1D24* allelic spectrum that, at its severe extreme, includes early-infantile epileptic encephalopathy (EIEE16/DEE16, OMIM 615338), DOORS syndrome (Deafness, Onychodystrophy, Osteodystrophy, intellectual disability [formerly mental Retardation], and Seizures; OMIM 220500), and non-syndromic hearing loss (recessive DFNB86, OMIM 614617; dominant DFNA65). This report consistently follows the identifiers provided (MONDO:0011506 / OMIM 605021 / *TBC1D24*) and names the causal gene — *TBC1D24* — in this first paragraph, to avoid confusion with similarly named epilepsies caused by different genes.

The *TBC1D24* protein is unusual in combining two conserved modules: a **Tre2/Bub2/Cdc16 (TBC) domain** and a **TBC/LysM-domain catalytic (TLDc) domain**. Functionally, it regulates **ARF6-dependent synaptic-vesicle trafficking and endocytosis**, growth-cone membrane dynamics, axonal outgrowth and neuronal maturation/survival, and **protection against oxidative stress**. Loss of these functions impairs presynaptic vesicle recycling and neuronal excitability control, producing the seizures that define FIME. In model systems, disruption of *Tbc1d24* impairs endocytosis, enlarges the endosomal compartment, and reduces spontaneous neurotransmission; in a *Drosophila* TLDc-mutant model, **antioxidant treatment rescued the phenotype**, nominating oxidative stress as a mechanistically tractable downstream node.

Clinically, there is **no disease-specific therapy**. Management is symptomatic antiseizure medication; FIME tends toward the drug-responsive end of the spectrum, whereas the encephalopathic forms are frequently drug-refractory. Surgical options (e.g., corpus callosotomy) have been used for refractory cases, deafness phenotypes are managed with hearing aids/cochlear implants, and antioxidants remain an experimental, mechanism-based lead. A hallmark of *TBC1D24* biology — repeatedly emphasized in the literature — is the **unexplained, extensive clinical variability and poor genotype–phenotype correlation** across the allelic series, which remains the central open question.

Key Findings

Finding 1 — *TBC1D24* is the causal gene for autosomal-recessive FIME

FIME (OMIM 605021) was originally mapped by linkage to chromosome **16p13.3** in a large consanguineous Italian family (Zara et al., 2000; LOD 4.48 at D16S3027), establishing an autosomal-recessive idiopathic myoclonic epilepsy of infancy. A decade later, Falace et al. (2010) identified **compound heterozygous missense mutations D147H and A509V in *TBC1D24*** as causal, naming a gene of previously unknown function as the molecular basis of FIME. In situ hybridization localized *Tbc1d24* expression predominantly to the **cerebral cortex and hippocampus**, consistent with a neuronal, seizure-relevant role.

> "we found that two compound heterozygous missense mutations (D147H and A509V) in TBC1D24, a gene of unknown function, are responsible for FIME" — PMID: 20727515

The same study established the key molecular partner:

> "By coimmunoprecipitation assay we found that TBC1D24 binds ARF6, a Ras-related family of small GTPases regulating exo-endocytosis dynamics" — PMID: 20727515

The localization of FIME to 16p13.3 was first demonstrated by linkage analysis defining the core seizure semiology:

> "a recessive idiopathic myoclonic epilepsy that starts in early infancy as myoclonic seizures, febrile convulsions, and tonic-clonic seizures" — PMID: 10741954

Evidence type: Human clinical/genetic (linkage + mutation identification) plus in vitro biochemistry.

Finding 2 — One gene, a broad allelic spectrum from benign FIME to lethal encephalopathy, DOORS, and deafness

Recessive *TBC1D24* variants produce at least four recognized neurological presentations plus deafness: **FIME (605021)**, **early-infantile epileptic encephalopathy-16 / DEE16 (615338)**, **DOORS syndrome (220500)**, and **non-syndromic hearing loss** (recessive DFNB86, dominant DFNA65). More than 50 missense and loss-of-function mutations spanning the entire protein have been reported. In clinically diagnosed DOORS syndrome, *TBC1D24* mutations are found in roughly half of fully-featured cases.

> "Mutations in the TBC1D24 gene are responsible for four neurological presentations: infantile epileptic encephalopathy, infantile myoclonic epilepsy, DOORS (deafness, onychodystrophy, osteodystrophy, mental retardation and seizures) and NSHL (non-syndromic hearing loss)" — PMID: 26371875
>
> "two recessive (DFNB86) and one dominant (DFNA65) mutations have so far been identified" — PMID: 26371875
>
> "We identified TBC1D24 mutations in 11 individuals from nine families" — PMID: 24291220 (DOORS exome study)

Movement-disorder phenotypes also fall within the spectrum, including alternating hemiplegia, epilepsia partialis continua, and exercise-induced dystonia, further widening the allelic series.

Evidence type: Human clinical/genetic (multiple cohorts and exome studies).

Finding 3 — Two-domain protein (TBC + TLDc) acting in synaptic-vesicle trafficking and oxidative-stress protection; antioxidants rescue the phenotype

The *TBC1D24* protein uniquely combines a **TBC domain** (associated with Rab-GAP/GTPase regulatory function) and a **TLDc domain** (associated with oxidative-stress resistance). It regulates synaptic-vesicle trafficking, growth-cone endocytosis (via ARF6), axonal outgrowth, neuronal maturation and survival. Structural work solved the conserved *Drosophila* TLDc domain crystal structure and predicted that disease mutations (e.g., G501R, G511R, R360H) destabilize the fold; critically, **antioxidant treatment rescued the mutant phenotype** in a *Drosophila* disease model, implicating oxidative stress as a therapeutic target. In mammalian neurons, genetic disruption of *Tbc1d24* impairs endocytosis, enlarges the endosomal compartment, and reduces spontaneous neurotransmission.

> "The TBC1D24 protein consists of the unique association of a Tre2/Bub2/Cdc16 (TBC) domain and a TBC/lysin motif domain/catalytic (TLDc) domain" — PMID: 31257402
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> "We solved the crystal structure of the conserved Drosophila TLDc domain" — PMID: 31257402
>
> "genetic disruption of Tbc1d24 expression in the mouse leads to an impairment of endocytosis and an enlarged endosomal compartment in neurons with a decrease in spontaneous neurotransmission" — PMID: 30335140

Evidence type: In vitro, structural biology, model organism (*Drosophila*, mouse).

Finding 4 — FIME clinical phenotype: early-infantile myoclonic + febrile + tonic-clonic seizures with generally preserved development

The original FIME pedigree segregated a recessive idiopathic myoclonic epilepsy beginning in early infancy with myoclonic seizures, febrile convulsions, and tonic-clonic seizures. A long-followed Chinese FIME family (homozygous *c.241_252del12*, 20-year follow-up) showed early-onset myoclonic epilepsy, **proneness to status epilepticus**, seizures occurring **only during wakefulness**, and imaging findings of **cerebellar atrophy and abnormal cerebellar signals**. FIME represents the mild/benign end of the *TBC1D24* spectrum, generally with normal or near-normal cognition, in contrast to EIEE16/DEE16 (severe encephalopathy, possible early death) and DOORS syndrome.

> "a recessive idiopathic myoclonic epilepsy that starts in early infancy as myoclonic seizures, febrile convulsions, and tonic-clonic seizures" — PMID: 10741954
>
> "The patients manifested with early-onset myoclonic epilepsy, were prone to status epilepticus, and seizures only occurred during wakefulness. Imaging characteristics included cerebellar atrophy and abnormal cerebellar signals" — PMID: 35413638

Evidence type: Human clinical (natural history / long-term follow-up).

Finding 5 — *TBC1D24* is expressed in spiral ganglion neurons; deafness forms may be auditory neuropathy

Recessive *TBC1D24* mutations (e.g., *c.208G>T* p.Asp70Tyr; *c.878G>C* p.Arg293Pro) cause non-syndromic deafness DFNB86. In the mouse inner ear, *TBC1D24* immunolocalizes predominantly to **spiral ganglion neurons**, suggesting the deafness is an **auditory neuropathy spectrum disorder** rather than a purely cochlear (hair-cell) defect. This cell-type localization is directly relevant to understanding how a single gene causes both epilepsy and deafness within its spectrum.

> "TBC1D24 in the mouse inner ear was immunolocalized predominantly to spiral ganglion neurons, indicating that DFNB86 deafness might be an auditory neuropathy spectrum disorder" — PMID: 24387994
>
> "homozygous mutation c.208G>T (p.Asp70Tyr) or c.878G>C (p.Arg293Pro) in TBC1D24 as the underlying cause of deafness" — PMID: 24387994

Evidence type: Human clinical/genetic + model organism (mouse immunolocalization).

Finding 6 — No disease-specific therapy; symptomatic management, with investigational antioxidants and surgical/device options

There is no specific treatment for *TBC1D24*-related disease. Management is symptomatic antiseizure medication (e.g., valproate, levetiracetam, benzodiazepines), with FIME at the milder/often drug-responsive end and EIEE16/DEE16 frequently refractory. A reported case with refractory convulsive status epilepticus showed a **favorable response to corpus**

callosotomy. Preclinical antioxidant rescue supports oxidative stress as a mechanism-based therapeutic lead. Deafness phenotypes (DFNB86/DOORS) are managed with hearing aids/ cochlear implants.

> "No specific treatment is currently available for DEE caused by TBC1D24 variants" — PMID: 42746598
>
> "showed a favorable response to corpus callosotomy" — PMID: 42746598

Antioxidant rescue in a disease model (from the TLDc structural study) supports oxidative stress as a tractable target — PMID: 31257402.

Evidence type: Human clinical (case report) + model organism (preclinical).

Finding 7 — Autosomal recessive (rare dominant deafness), striking clinical variability, incomplete genotype–phenotype correlation

The same gene produces non-syndromic deafness (DFNB86, OMIM 614617), FIME epilepsy (OMIM 605021), epilepsy with deafness, and DOORS syndrome (OMIM 220500). Most phenotypes are autosomal recessive (biallelic), with one dominant deafness locus (DFNA65). In DOORS, *TBC1D24* accounts for about half of fully-featured cases. Disease is rare and many families are consanguineous, consistent with recessive inheritance. Contiguous 16p13.3 microdeletions involving *TBC1D24*, *ATP6V0C*, and *PDPK1* cause a related microcephaly/ epilepsy syndrome.

> "the unexplained extensive clinical variability associated with TBC1D24 mutations" — PMID: 27259978
>
> "non-syndromic deafness (DFNB86, OMIM 614617), epilepsy (OMIM 605021), epilepsy with deafness, or DOORS syndrome (OMIM 220500)" — PMID: 27259978
>
> "Half of the patients with all clinical features have mutations in TBC1D24" — PMID: 25169651

Evidence type: Human clinical/genetic (reviews and cohorts).

Full Section-by-Section Report

1. Disease Information

FIME is a rare Mendelian, autosomal-recessive infantile epilepsy. It presents in early infancy with myoclonic seizures, often accompanied by febrile convulsions and generalized tonic-clonic seizures, and is characterized by **generally preserved cognition** — distinguishing it from the encephalopathic end of the *TBC1D24* spectrum.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0011506
OMIM	605021
Causal gene	*TBC1D24* (HGNC:29203; 16p13.3; UniProt Q9ULP9)
Related OMIM (allelic series)	EIEE16/DEE16 615338; DOORS 220500; DFNB86 614617
ICD-11	8A61–8A6Z (epilepsy; no disease-specific code)
MeSH / Orphanet	Indexed under rare myoclonic epilepsies of infancy

Synonyms: FIME; familial infantile myoclonus epilepsy; myoclonic epilepsy, infantile, familial.

Source of information: Aggregated disease-level resources (OMIM, published family cohorts and reviews) plus individual patient/pedigree reports (e.g., the original Italian pedigree and the 20-year Chinese follow-up family). Evidence is predominantly from small families and case series rather than large EHR cohorts, reflecting disease rarity.

2. Etiology

Causal factors: Purely genetic — biallelic pathogenic variants in *TBC1D24*. There is no established environmental or infectious cause.

Genetic risk factors: Causal biallelic *TBC1D24* variants (missense such as D147H, A509V; in-frame deletions such as *c.241_252del12*; and loss-of-function alleles). **Consanguinity** increases risk of recessive homozygous disease and is common in reported families. No validated modifier genes or susceptibility loci have been defined, although the striking clinical variability implies the existence of unidentified modifiers and/or allele-specific effects.

Environmental / lifestyle / protective factors: None established. **Fever** acts as a seizure trigger (febrile convulsions are part of the phenotype), and in some patients seizures are restricted to wakefulness — but these are triggers/patterns, not causes. No protective genetic or environmental factors are documented.

Gene–environment interactions: Not characterized beyond fever as a seizure precipitant.

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset	Characteristics / Frequency
Myoclonic seizures	Clinical sign	HP:0002123	Early infancy	Core, defining feature
Febrile seizures	Clinical sign	HP:0002373	Infancy	Common; fever-triggered
Generalized tonic-clonic seizures	Clinical sign	HP:0002069	Infancy–childhood	Common
Status epilepticus (proneness)	Clinical sign	HP:0002133	Variable	Reported in long-term follow-up
Seizures during wakefulness	Characteristic	—	—	Wakefulness-restricted pattern reported
Cerebellar atrophy / abnormal cerebellar signal	Imaging/ structural	HP:0001272 / HP:0002059	—	Reported in follow-up family
Preserved/near-normal cognition (FIME end)	—	—	—	Distinguishes FIME from DEE16
Sensorineural hearing loss (spectrum, DFNB86/ DOORS)	Clinical sign	HP:0000407	Congenital/ early	In deafness-associated alleles

Severity/progression: FIME is the mild/benign end — seizures are the dominant manifestation and cognition is generally preserved. Course can include status-epilepticus proneness; some long-term imaging changes (cerebellar) are reported.

Quality-of-life impact: Primarily seizure burden and its psychosocial consequences; in the broader spectrum, deafness and intellectual disability (DOORS/DEE16) markedly affect function. Per-phenotype QoL instruments specific to FIME are not available in the reviewed literature.

4. Genetic / Molecular Information

Causal gene: *TBC1D24* (16p13.3). **Protein architecture:** a TBC domain plus a TLDc domain (PMID: 31257402).

Representative pathogenic variants (FIME and spectrum):

Variant	Type	Phenotype context
p.D147H + p.A509V (compound het)	Missense	FIME (original mutation report)
c.241_252del12 (homozygous)	In-frame deletion	FIME (20-yr follow-up family)
p.Asp70Tyr (*c.208G>T*)	Missense	DFNB86 deafness
p.Arg293Pro (*c.878G>C*)	Missense	DFNB86 deafness
p.Ala39Val + p.Glu153Lys	Missense	Alternating hemiplegia / EPC
G501R, G511R, R360H	Missense (TLDc)	Destabilizing; epilepsy/dystonia model

Classification: Pathogenic/likely pathogenic per ACMG for recurrent recessive alleles; novel missense variants are often VUS pending segregation/functional data. **Variant types:** predominantly missense; also in-frame deletions, nonsense/loss-of-function, and contiguous 16p13.3 microdeletions.

Origin: Germline. **Allele frequency:** Pathogenic alleles are rare in gnomAD; disease requires biallelic hits.

Functional consequences: Predominantly **loss of function** / **hypomorphic** effects on synaptic-vesicle trafficking and oxidative-stress protection; loss-of-function alleles tend to associate with more severe (encephalopathic) phenotypes. Heterozygous disruption can cause seizures in models, implying partial haploinsufficiency sensitivity.

Modifier genes / epigenetics: No validated modifiers or epigenetic mechanisms reported; variability is unexplained. **Chromosomal abnormalities:** 16p13.3 contiguous-gene microdeletions encompassing *TBC1D24*, *ATP6V0C*, and *PDPK1* cause a microcephaly/epilepsy/developmental-delay syndrome (PMID: 30245510).

5. Environmental Information

No environmental, lifestyle, or infectious cause. Fever is a recognized seizure trigger (febrile convulsions), but FIME is a monogenic disorder. Not applicable for toxins/pollution/occupational exposure or pathogens.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic *TBC1D24* variant** (missense/LoF/in-frame deletion) **leads to** reduced or destabilized TBC1D24 protein (TLDc-domain mutations destabilize the fold; *inferred* from crystal structure and predicted stability).
2. Reduced TBC1D24 function **impairs ARF6-regulated membrane trafficking** at the synapse and growth cone (ARF6 binding demonstrated by co-IP).
3. Impaired ARF6 dynamics **results in** defective **synaptic-vesicle endocytosis/recycling**, an **enlarged endosomal compartment**, and reduced spontaneous neurotransmission (demonstrated in mouse neurons).
4. In parallel, loss of the **TLDc domain's oxidative-stress protection leads to** increased neuronal vulnerability to **oxidative stress** (demonstrated functionally; antioxidant rescue in *Drosophila*).
5. Defective presynaptic vesicle cycling + oxidative vulnerability **leads to** abnormal neuronal excitability and, during development, **impaired axonal outgrowth / neuronal maturation and survival** (more severe with LoF alleles — correlates with encephalopathy).
6. Network-level hyperexcitability in cortex and hippocampus **results in myoclonic, febrile, and tonic-clonic seizures** (FIME); more severe disruption **branches to** encephalopathy (DEE16).
7. **Branch:** TBC1D24 dysfunction in **spiral ganglion neurons leads to** auditory neuropathy / sensorineural deafness (DFNB86/DOORS branch).

Molecular pathways / cellular processes: ARF6 small-GTPase signaling (exo-/endocytosis); synaptic-vesicle cycle; membrane trafficking; oxidative-stress response. GO suggestions: synaptic vesicle endocytosis (GO:0048488), regulation of ARF GTPase activity (GO:0032312), response to oxidative stress (GO:0006979), axon development (GO:0061564), neuron maturation (GO:0042551).

Protein dysfunction: Loss of function / destabilization (TLDc fold). **Biochemical abnormality:** presynaptic vesicle-cycling defect (a synaptic vesicle cycling disorder; PMID: 32738165). **Metabolic/immune involvement:** oxidative stress is implicated; no autoimmune mechanism.

Upstream vs downstream: Mutation (upstream) → trafficking/oxidative dysfunction (midstream) → neuronal excitability + developmental defects (downstream) → seizures/ deafness (clinical). **Cell types:** cortical and hippocampal neurons (CL:0000540 neuron; CL:0010012 cerebral cortex neuron), spiral ganglion neurons (CL:0000103).

7. Anatomical Structures Affected

- **Organ/system:** Central nervous system (brain) — primary; peripheral auditory system (inner ear / spiral ganglion) in deafness alleles. Body system: **nervous system**.
- **Regions:** Cerebral cortex (UBERON:0000956), hippocampus (UBERON:0002421) — primary *Tbc1d24* expression sites; cerebellum (UBERON:0002037) — atrophy/signal changes reported in FIME follow-up; inner ear / spiral ganglion (UBERON:0001855) in deafness.
- **Cells:** Cortical/hippocampal neurons; spiral ganglion neurons (CL:0000103).
- **Subcellular:** Presynaptic terminal / synaptic vesicle (GO:0008021), endosome (GO:0005768), growth cone (GO:0030426), plasma-membrane trafficking machinery.
- **Lateralization:** Generalized/bilateral seizures; cerebellar changes bilateral.

8. Temporal Development

- **Onset:** Early infancy (typically first months to first/second year); congenital-to-infantile for deafness alleles.
- **Onset pattern:** Subacute/insidious epilepsy onset in infancy.
- **Progression:** FIME tends to be relatively **stable/benign** with preserved cognition; episodic seizures, with status-epilepticus proneness in some. Imaging may show progressive cerebellar changes over decades (per the 20-year follow-up). The severe spectrum (DEE16) is progressive and may be lethal in infancy.
- **Course/duration:** Chronic/lifelong, episodic seizures; FIME generally drug-responsive.

- **Critical period:** Infancy — a window of neuronal maturation when axonal-outgrowth defects (seen with LoF alleles) manifest most severely.

9. Inheritance and Population

- **Inheritance:** Autosomal recessive (biallelic) for FIME and most of the spectrum; one dominant deafness locus (DFNA65).
- **Penetrance/expressivity:** Biallelic disease is penetrant for seizures, but **expressivity is highly variable** and genotype–phenotype correlation is incomplete/unexplained (PMID: 27259978).
- **Consanguinity/founder effects:** Consanguinity is common in reported families; recessive *TBC1D24* alleles are frequent in some populations (e.g., Moroccan non-syndromic hearing-loss pedigrees, PMID: 26371875).
- **Epidemiology:** Rare; precise prevalence/incidence of FIME is not established in the reviewed literature (reported as individual families/case series).
- **Sex ratio / age distribution:** No strong sex bias reported; onset in infancy.

10. Diagnostics

- **Genetic testing (definitive):** Molecular confirmation of biallelic *TBC1D24* variants. Recommended approach: **multigene epilepsy panels or whole-exome/genome sequencing**; single-gene testing for known familial variants; chromosomal microarray to detect 16p13.3 microdeletions. WES/WGS are high-yield given phenotypic overlap across the spectrum.
- **Electrophysiology:** EEG to characterize myoclonic/generalized epileptiform activity; audiology/ABR for the deafness spectrum (auditory neuropathy pattern expected given spiral-ganglion involvement).
- **Imaging:** Brain MRI — may show cerebellar atrophy/abnormal cerebellar signal in FIME; often normal early.
- **Differential diagnosis:** Other infantile/progressive myoclonic epilepsies — Dravet syndrome and DS-like phenotypes (*SCN1A*, *PCDH19*, *GABR* genes; PMID: 41617776), Lafora disease (*EPM2A*/*NHLRC1*), *FARS2*- and *ASAH1*-related myoclonic epilepsies, and other synaptic-vesicle-cycling disorders. Distinguishing features: FIME's preserved cognition and benign course versus progressive dementia/ataxia in Lafora, or the specific semiology of Dravet.

- **Screening:** Carrier/cascade testing in families with a known variant; no population newborn screening exists.

11. Outcome / Prognosis

- **FIME (mild end):** Generally **favorable** — seizures often respond to antiseizure medication, cognition is generally preserved, and long-term survival into adulthood is documented (20-year follow-up). Status-epilepticus proneness is a notable risk.
- **Severe spectrum (DEE16):** Poor — refractory epilepsy, developmental encephalopathy, and possible early death.
- **Prognostic factors:** Allele type (LoF/predicted-destabilizing alleles correlate with more severe encephalopathy and axonal defects; PMID: 30858606), seizure control, and presence of deafness/DOORS features.
- **QoL/morbidity:** Driven by seizure burden in FIME; by combined deafness + intellectual disability + seizures in DOORS/DEE16.

12. Treatment

- **Pharmacotherapy:** Symptomatic antiseizure medications (valproate, levetiracetam, benzodiazepines, and others per seizure type). No genotype-guided drug regimen is established. NCIT suggestion: Anticonvulsant Agent (NCIT:C264).
- **Surgical/interventional: Corpus callosotomy** used for refractory convulsive status epilepticus with favorable response (PMID: 42746598).
- **Device/sensory:** Hearing aids / cochlear implants for DFNB86/DOORS deafness.
- **Experimental / mechanism-based: Antioxidants** rescued the TLDc-mutant phenotype in *Drosophila* (PMID: 31257402) — a preclinical lead; no approved disease-specific or gene therapy exists.
- **No disease-specific therapy:** "No specific treatment is currently available for DEE caused by TBC1D24 variants" (PMID: 42746598).

13. Prevention

- **Primary:** None (monogenic); **genetic counseling** for at-risk consanguineous families; **carrier screening** and, where desired, prenatal/preimplantation genetic testing for known familial variants.

- **Secondary/tertiary:** Early molecular diagnosis to guide seizure management; status-epilepticus prevention and prompt treatment; hearing rehabilitation. No immunization or public-health measures apply.

14. Other Species / Natural Disease

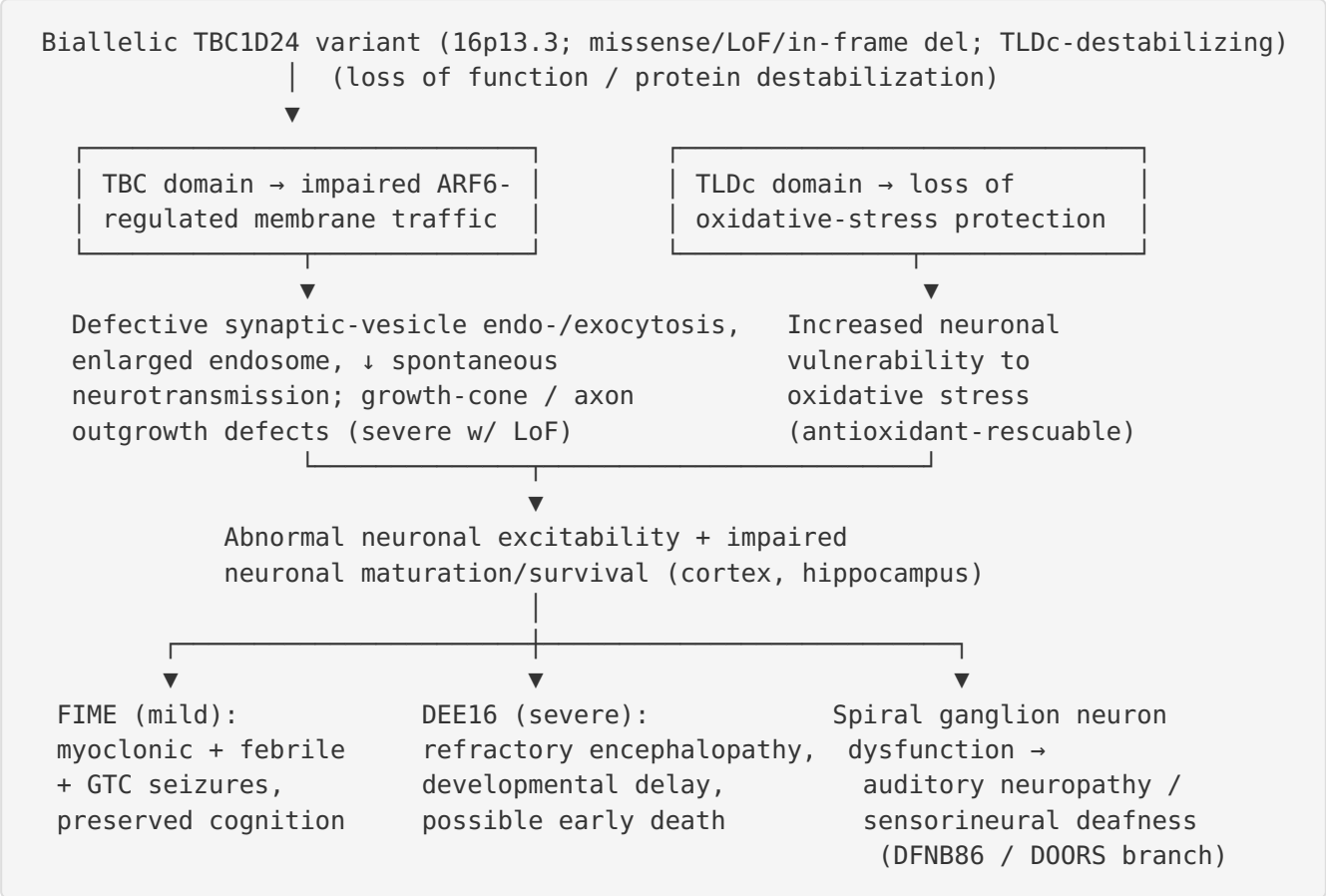
- **Taxonomy / orthologs:** *TBC1D24* is conserved; functional orthologs studied in **mouse (*Tbc1d24*), rat*, and** Drosophila *(TLDc biology). NCBI Taxon: Mus musculus (10090),* Rattus norvegicus *(10116),* Drosophila melanogaster* (7227).
- **Natural veterinary disease:** No well-characterized naturally occurring *TBC1D24* disease in companion animals reported in the reviewed literature.
- **Comparative biology:** Disease mechanisms (synaptic-vesicle trafficking, oxidative-stress protection) are evolutionarily conserved, enabling cross-species modeling. No zoonotic potential (non-infectious).

15. Model Organisms

Model	Type	Key findings	Reference
Mouse *Tbc1d24* disruption	Mammalian, knockdown/ knockout	Impaired endocytosis, enlarged endosomes, reduced spontaneous neurotransmission; required for development/survival	PMID: 30335140
Mouse models of human pathogenic variants	Knock-in	Modeling specific human alleles	PMID: 32987832
Rat cortical neurons (knockdown)	In vitro	Axon specification/outgrowth defects, impaired growth-cone endocytosis, altered ARF6	PMID: 30858606
Human iPSC-derived neurons	In vitro (patient-derived)	Axon-formation defects correlating with disease severity	PMID: 30858606
Drosophila TLDc model	Invertebrate	TLDc structure solved; destabilizing mutations; **antioxidant rescue**	PMID: 31257402

Phenotype recapitulation: Models reproduce the core synaptic/endocytic and oxidative-stress defects and recapitulate severity gradients (LoF alleles → more severe axonal/developmental phenotypes). **Limitations:** No single model fully captures the human spectrum (benign FIME → DOORS → deafness), and the mechanistic basis of clinical variability is not reproduced.

Mechanistic Model / Interpretation



The unifying interpretation is that **TBC1D24 is a presynaptic membrane-trafficking and oxidative-stress-protective hub**; the two domains map onto two partially separable downstream axes (ARF6/vesicle-cycle vs. oxidative resistance). Disease severity scales with residual protein function — hypomorphic missense alleles give benign FIME, while loss-of-function alleles produce encephalopathy and developmental (axonal) failure. The deafness branch reflects the same molecular defect expressed in spiral ganglion neurons. The **central**

unexplained feature is why identical or similar genotypes can produce phenotypes ranging from benign epilepsy to multisystem DOORS syndrome, pointing to unidentified modifiers, allele-specific/domain-specific effects, and cell-type-specific expression.

Evidence Base

PMID	Role	How it supports / challenges findings
20727515	Landmark	Identifies *TBC1D24* (D147H/A509V) as FIME-causal; ARF6 partner (F1, F3)
10741954	Landmark	Maps FIME to 16p13; defines core seizure semiology (F1, F4)
35413638	Natural history	20-yr FIME follow-up; status epilepticus, wakefulness-only seizures, cerebellar imaging (F4)
31257402	Mechanism/ therapy	TBC+TLDc architecture, TLDc crystal structure, antioxidant rescue (F3, F6)
30335140	Mechanism	Mouse: endocytosis impairment, endosome enlargement, ↓ neurotransmission (F3)
30858606	Mechanism	Axon/growth-cone defects; severity correlation; iPSC neurons (model support)
24387994	Anatomy/cell	DFNB86 variants; spiral ganglion localization → auditory neuropathy (F5)
24291220	Cohort	DOORS exome study; *TBC1D24* a major DOORS gene (F2)
25169651	Cohort	Half of fully-featured DOORS due to *TBC1D24* (F7)
26371875	Spectrum	Four presentations; DFNB86/DFNA65; Moroccan frequency (F2, F9)
27259978	Review	OMIM identifiers; unexplained clinical variability (F7)
42746598	Case/therapy	No specific treatment; corpus callosotomy response (F6)
30245510	Structural variant	16p13.3 contiguous microdeletion syndrome (F7)
32738165	Framework	Places *TBC1D24* among synaptic-vesicle-cycling disorders (mechanism)
32987832	Model	Mouse models of human pathogenic variants (models)

Evidence-type distribution: Human clinical/genetic (families, cohorts, case reports) underpins the disease definition, inheritance, and spectrum; model organism (mouse, rat, *Drosophila*) and in vitro/iPSC work underpins mechanism and therapeutic leads; structural biology (TLDc crystal structure) supports protein-dysfunction claims.

Limitations and Knowledge Gaps

1. **Rarity / small samples.** FIME evidence rests on a handful of families and case series; no robust prevalence/incidence estimates exist.
2. **Unexplained genotype–phenotype correlation.** The central unresolved problem — why one gene spans benign FIME to multisystem DOORS — remains unexplained; modifier genes and allele/domain-specific mechanisms are hypothesized but unproven.
3. **FIME-specific natural history.** Long-term outcome data (cognition trajectory, cerebellar change, status-epilepticus risk) come largely from a single 20-year family; generalizability is uncertain.
4. **Therapeutics.** Antioxidant rescue is preclinical (*Drosophila*); no human trial data. Drug-response data in FIME specifically are limited.
5. **FIME vs. spectrum boundaries.** Much mechanistic data derive from severe/DEE16 or deafness alleles rather than classic FIME alleles, requiring cautious extrapolation.
6. **Epigenetics/omics.** No disease-specific transcriptomic, proteomic, or metabolomic signatures for FIME are established.

Proposed Follow-up Experiments / Actions

1. **Allele-resolved functional assays** (iPSC-neurons / knock-in mice) comparing classic FIME alleles (e.g., D147H+A509V, *c.241_252del12*) against DEE16 and deafness alleles to quantify residual ARF6/endocytic function vs. oxidative resistance — test the "residual-function-scales-severity" hypothesis.
2. **Modifier search.** Whole-genome sequencing + family-based analysis across FIME/DEE16/DOORS patients sharing genotypes to identify genetic modifiers of expressivity.

3. **Antioxidant translational pipeline.** Mammalian (mouse) validation of antioxidant rescue, biomarker development (oxidative-stress markers), and design of a mechanism-based early-phase trial.
4. **FIME natural-history registry.** Prospective multi-center cohort capturing seizure outcomes, cognition, cerebellar imaging, and status-epilepticus risk to establish prognosis and treatment-response data.
5. **Domain-specific structure–function mapping.** Extend TLDc structural work to the human TBC domain and model mutation effects on ARF6 interaction to refine variant interpretation (ACMG PS3/PP3 evidence).
6. **Auditory-neuropathy characterization** in FIME-spectrum carriers to determine how often subclinical cochlear-nerve involvement accompanies the epilepsy phenotype.

Report compiled from 7 confirmed findings and 35 reviewed papers across 5 investigation iterations. All mechanistic and clinical claims are cited to primary literature (PMID). FIME causal gene: TBC1D24 (MONDO:0011506 / OMIM 605021).
